Cancer Variant Analysis

Introduction to Cancer Genomics and Variant Detection

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Learning Objectives

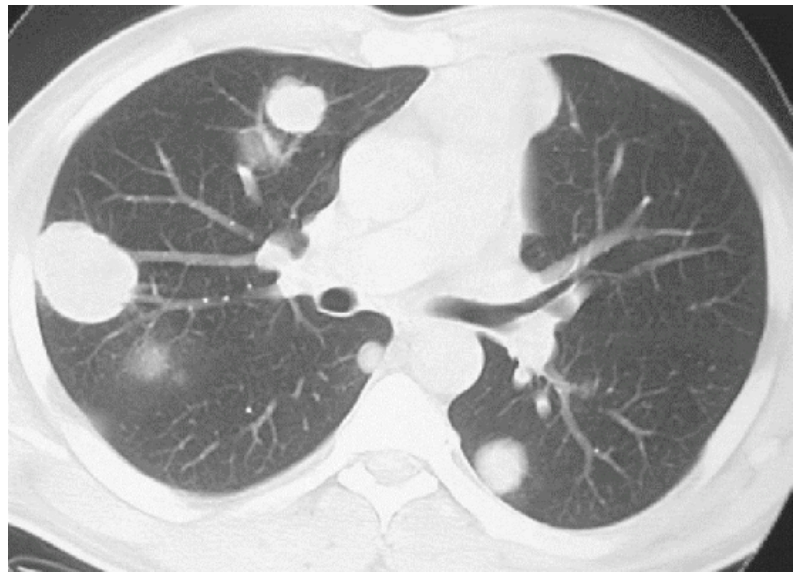
After this session, you will be able to:

- Understand why cancer is fundamentally a genomic disease
- Distinguish between somatic and germline variants
- Identify different types of mutations in cancer
- Understand sequencing approaches for cancer genomics
- Comprehend the bioinformatics workflow for variant detection
- Appreciate the clinical relevance of variant calling

Cancer: A Disease of the Genome

Key characteristics:

- **Abnormal cell growth** - uncontrolled proliferation
- **Invasive potential** - ability to spread (metastasis)
- Initiated by **acquired genomic mutations** affecting cell growth regulators
- Mutations occur **stochastically**, but rate influenced by environment
- **Clonal evolution** - natural selection of malignant cells



CT image showing multiple lung masses in a patient with metastatic testicular seminoma, Source

Driver vs Passenger Mutations

Cancer genomes typically harbor **2-8 driver mutations** (Vogelstein et al., 2013), though this varies by cancer type. Pediatric cancers often have fewer drivers, while hypermutated adult cancers may have more. Drivers represent a small minority of total mutations.

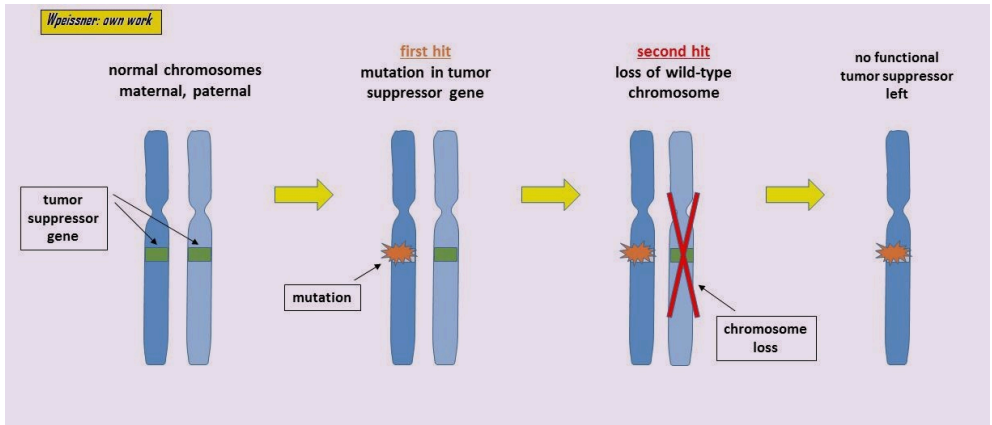
Hereditary vs Sporadic Cancer

Sporadic (~90-95%)

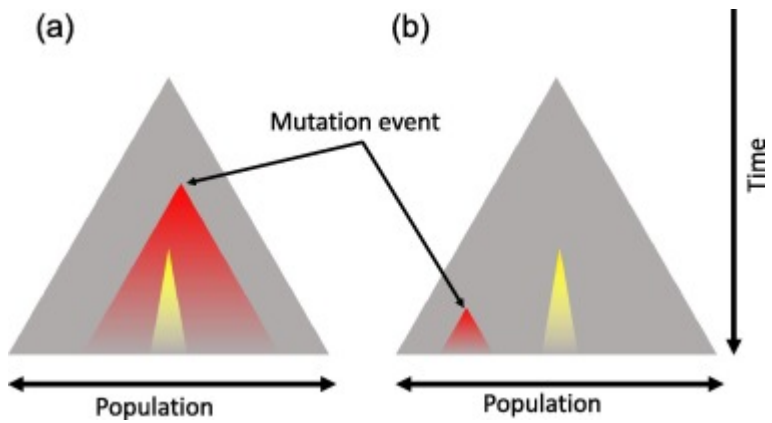
- Acquired somatic mutations
- Accumulate during lifetime
- Environmental factors contribute

Hereditary (~5-10%)

- Inherited predisposition alleles
- **BRCA1/2** (breast, ovarian)
- **Lynch syndrome** genes (colorectal)
- Still require somatic “second hits”



Knudson hypothesis: Tumor Suppressor Gene (Knudson, 1971), Source



Timing of Mutation Events: An earlier mutation (a) produces a larger population of mutant cells, than a later event (b), (Oota, 2020) Source

Clinical Relevance

Hereditary cancer syndromes have implications for family screening, risk reduction, and treatment options (e.g., PARP inhibitors for BRCA carriers).

The Hallmarks of Cancer

8 Core Hallmarks:

Hallmark	Description
Sustaining proliferative signaling	Self-sufficiency in growth signals
Evading growth suppressors	Insensitivity to anti-growth signals
Resisting cell death	Evading apoptosis
Enabling replicative immortality	Limitless replicative potential
Inducing angiogenesis	Sustained blood vessel formation
Activating invasion & metastasis	Tissue invasion and spread
Avoiding immune destruction	Escaping immune surveillance
Deregulating cellular energetics	Altered metabolism (Warburg effect)

2 Enabling Characteristics: Genome instability & mutation; Tumor-promoting inflammation

Reference

Hanahan & Weinberg (2011) “Hallmarks of Cancer: The Next Generation” - Cell

Variant Terminology: Essential Definitions

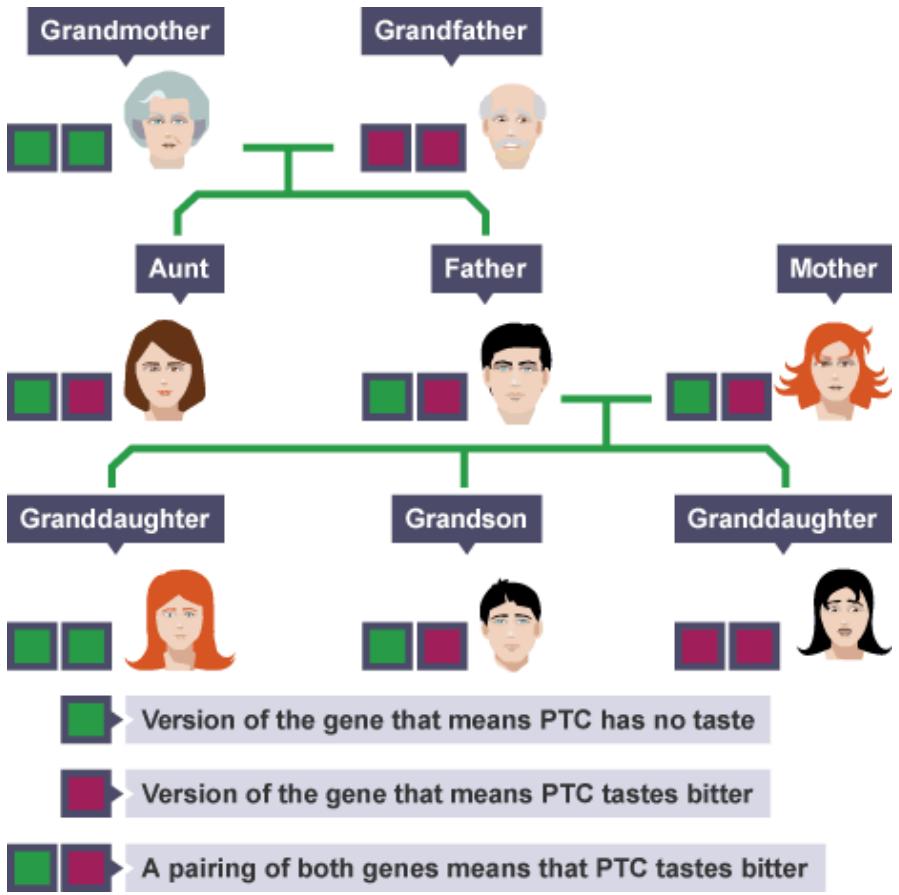
Mutation: The *process* of change in DNA

Variant: A difference in DNA sequence compared to a reference (the **outcome**)

Somatic variant: Occurs only in specific cells/tissues - Not inherited - Arises during lifetime

Germline variant: Present in all cells - Can be passed to offspring - Present from conception

Polymorphism: Traditionally defined as variant >1% frequency in population (though “variant” is now preferred terminology)



PTC (Phenylthiocarbamide): A chemical that tastes **bitter** only to those with a specific inherited dominant variant. This pedigree shows its germline inheritance pattern. Source

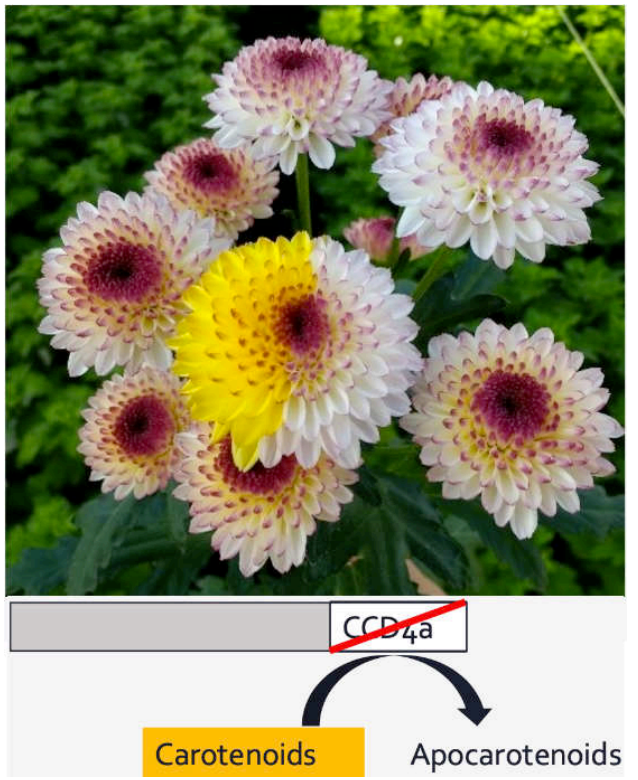
Mutation vs Variant: A Practical Example

The Yellow Flower Example:

- **Mutation**: The change in DNA that caused petals to turn yellow
- **Variant**: The resulting DNA difference between yellow and white flowers

This is a **somatic mutation** - occurring during the plant's development in one branch, analogous to somatic mutations in cancer.

The CCD4a gene mutation prevents breakdown of carotenoids, leading to yellow pigmentation.



Mutation example: chrysanthemum flowers (credit: Geert van Gest). Source

Types of Mutations in Cancer

Small-scale mutations

- **SNVs** (Single Nucleotide Variants)
- **INDELs** (Insertions and Deletions)

Structural variations

- **Large INDELs** (>50bp)
- **Translocations** - chromosomal rearrangements
- **Inversions** - reversed DNA segments
- **Fusion transcripts** - gene fusions from translocations
- **CNV** (Copy Number Variation) - gains/losses of genomic regions

LOH (Loss of Heterozygosity) is a *consequence* of CNV or copy-neutral events, not a mutation type itself.

> **Clinical Relevance**
>
> Different mutation types require different detection methods and have distinct clinical implications for treatment selection.

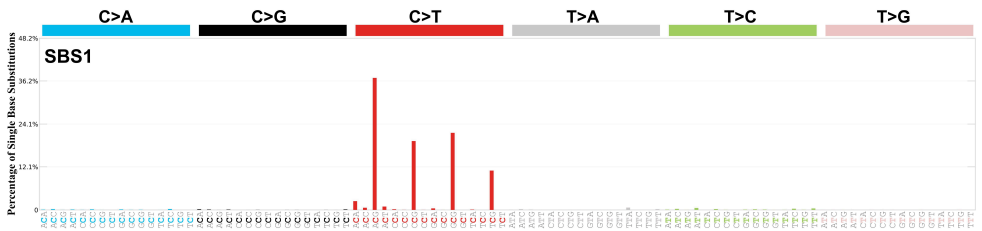
Mutational Signatures: Fingerprints of Mutagenesis

Each mutational process leaves a characteristic pattern:

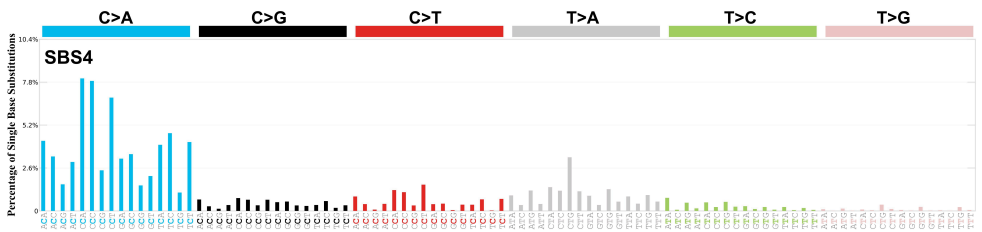
- **Signature 1**: Aging, C>T at CpG sites, **Transition** (Py ↔ Py)
- **Signature 4**: Tobacco smoking, C>A, **Transversion** (Py ↔ Pu)
- **Signature 6**: Mismatch repair (MMR) deficiency
- **Signature 7**: UV light, C>T at dipyrimidines, **Transition**

Clinical applications:

- Understanding tumor etiology (e.g. Signature 4, 7)
- Treatment selection (e.g., PARP inhibitors for HRD) (Signature 3)
- Immunotherapy response prediction (Signature 6)



Mutational signature SBS1: Ageing signature. Source



Mutational signature SBS4: Smoking signature. Source

Transition: Swapping similar shapes (Purine ↔ Purine or Pyrimidine ↔ Pyrimidine).
Transversion: Swapping different shapes (Purine ↔ Pyrimidine).

Reference

Alexandrov et al. (2020) “The repertoire of mutational signatures in human cancer” - Nature

Sequencing Strategies for Cancer Genomics

Coverage strategies

- **Whole Genome Sequencing (WGS)**
  - Complete genome coverage
  - Detects all variant types
  - Best for structural variants
- **Whole Exome Sequencing (WES) (Bait Capture)**
  - Protein-coding regions only
  - Cost-effective (
    WES 100x: 25 M 2 x 100 bp
    WGS 30x: 450 M 2 x 100 bp
)
  - Misses non-coding regions
- **Custom panels (Bait Capture, mostly)**
  - Targeted cancer genes
  - Deep coverage

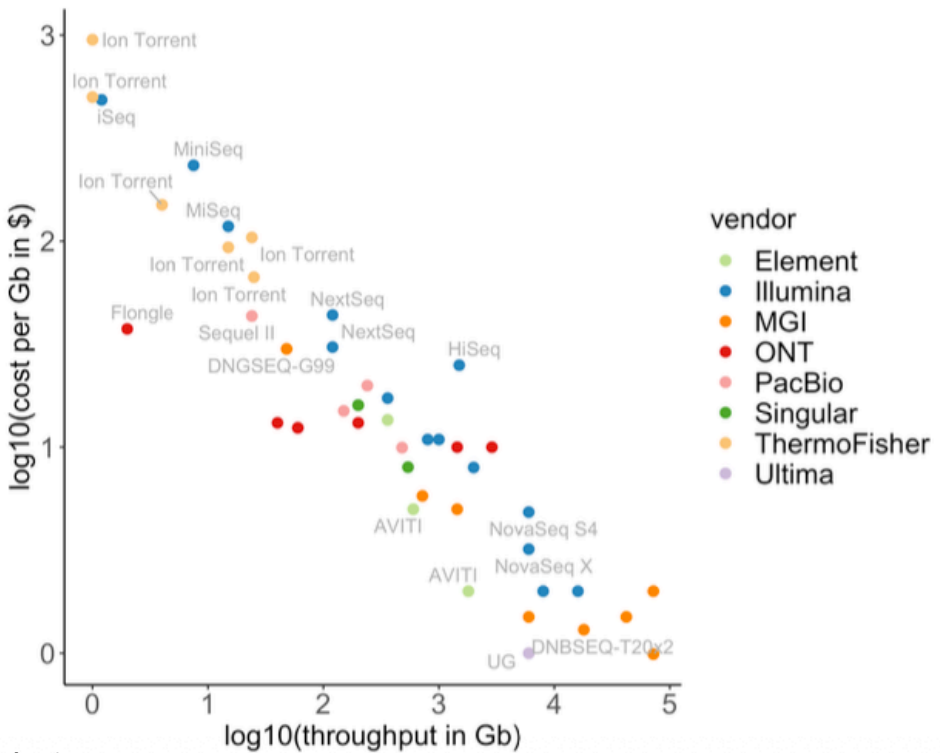
Sequencing technologies

Short reads (2x150bp standard):

- Illumina, MGI, Element, Ultima
- High accuracy, mature pipelines

Long reads:

- PacBio HiFi: >Q30 accuracy
- Oxford Nanopore: Q20+ with R10.4.1
- Superior for SVs and phasing



Sequencing cost vs throughput, credit: Geert van Geest

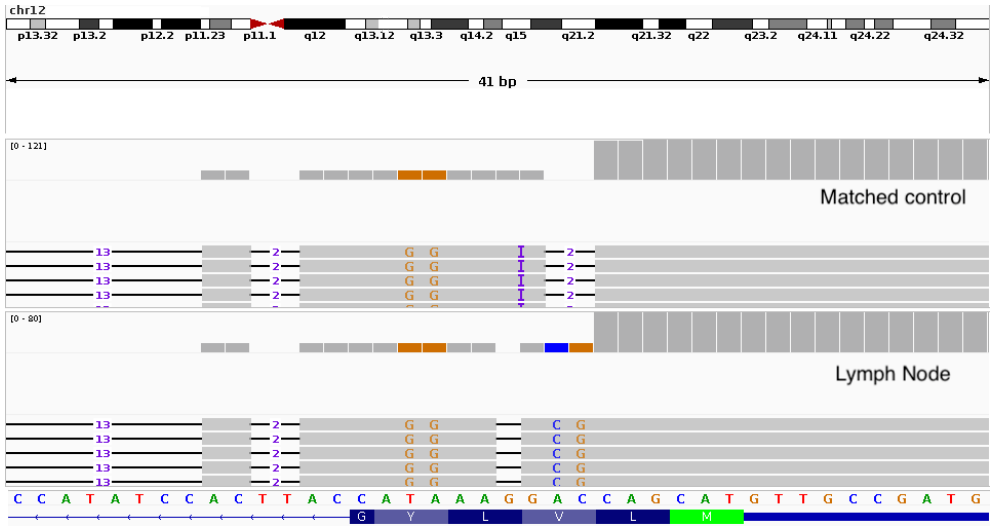
Experimental Design: Tumor-Normal Pairs

The challenge: Tumor tissue is heterogeneous - contains: - Tumor cells - Immune infiltrates - Stromal cells - Normal tissue

Solution: Paired samples

1. **Tumor sample** - from the malignancy
2. **Normal sample** - typically blood

For hematological malignancies, use skin biopsy or buccal swab (blood IS the tumor!)



IGV normal matched vs tumor

> **Tumor Purity**
>
> Typical tumor purity ranges from 30-80%. **Samples below 20-30% purity may have insufficient power for reliable somatic variant detection.** Consider pathologist review or microdissection.

Bioinformatics Workflow Overview



Critical Note

Variant **annotation** is essential but often overlooked! Without functional annotation and clinical database lookup, variants have limited utility.

Note

Course Scope

The steps highlighted in yellow are directly discussed in this course.

The remaining steps are not mentioned here but are covered in other courses (e.g., NGS - Variant Analysis and NGS - Quality Control, Alignment, Visualisation). However, we provide the scripts for the analysis in the course GitHub repo.

The SAM/BAM/CRAM File Formats

Purpose: Store sequence alignments

SAM: Text-based (Human readable)

BAM: **Lossless** binary compressed SAM

- *Indexable (Fast random access)*

CRAM: **Ref-based** compression

- *Lossless or Lossy modes*
- ~30-60% smaller than BAM
- Requires access to the reference genome for decoding

Example from our BAM files

Commands from the quality control exercises, run on `normal.recal.bam`. `samtools view` prints the BAM file as SAM text:

1. Header (`@` lines, shortened: 6 of 9 lines)

```
1  ALIGNDIR=~/project/data/alignments
2  samtools view -H "$ALIGNDIR"/normal.recal.bam
```

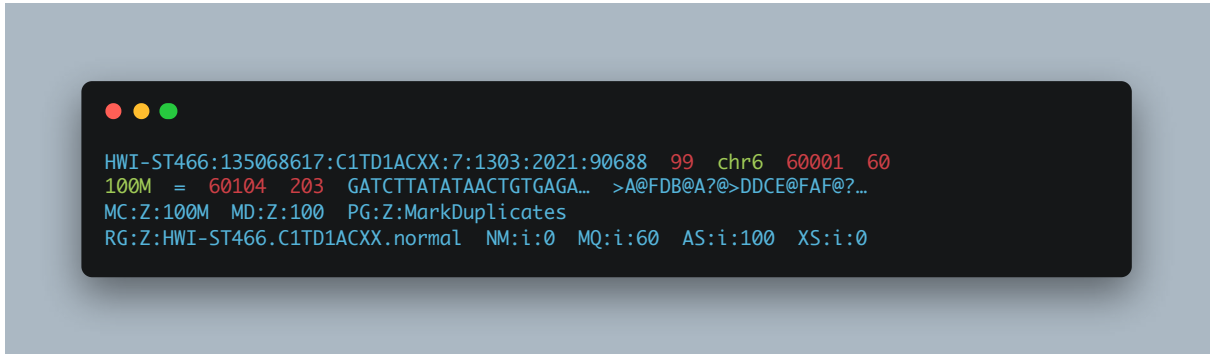
Line	Fields	Our file
`@HD`	`VN` format version, `SO` sort order	`1.6`, `coordinate`
`@SQ`	`SN` reference name, `LN` length	`chr6` 170,805,979 bp, `chr17` 83,257,441 bp
`@RG`	`ID` group, `SM` sample, `LB` library, `PL` platform, `PU` flowcell	`normal`, `ILLUMINA`, `HWI-ST466.C1TD1ACXX`
`@PG`	`ID` `PN` program, `VN` version, `PP` previous program, `CL` command	`bwa` 0.7.19, `samtools` 1.22.1

```
@HD  VN:1.6  SO:coordinate
@SQ  SN:chr6  LN:170805979
@SQ  SN:chr17  LN:83257441
@RG  ID:HWI-ST466.C1TD1ACXX.normal  LB:normal  PL:ILLUMINA  SM:normal
PU:HWI-ST466.C1TD1ACXX
@PG  ID:bwa  PN:bwa  VN:0.7.19-r1273  CL:bwa mem /config/project/course_data
/reference/ref_genome.fa /config/project/course_data/reads/normal_R1.fastq.gz /config
/project/course_data/reads/normal_R2.fastq.gz
@PG  ID:samtools  PN:samtools  PP:bwa  VN:1.22.1  CL:samtools sort -o /config/project
/course_data/alignments/normal.bam -
```

2. Alignment Record (The actual reads; shown: the second read of the output, wrapped, SEQ and QUAL shortened with …)

```
samtools view "$ALIGNDIR"/normal.recal.bam | head -3
```

Field	Meaning
QNAME	Query / Read Name (HWI-ST466:135068617:C1TD1ACXX:7:1303:2021:90688)
FLAG	Bitwise flags describing state (99)
RNAME	Reference sequence / Chromosome (chr6)
POS	1-based mapping position (60001)
MAPQ	Mapping quality / Confidence (60)
CIGAR	Alignment String (100M)
SEQ	Nucleotide sequence (GATCTTATATAACTGTGAGA…)
QUAL	Base quality scores (>A@FDB@A?@>DDCE@FAF@?…)



Additional fields of the alignment record (the same in SAM and BAM)

- **RNEXT** = = (mate maps to same chromosome)
- **PNEXT** = 60104 (mate's mapping position)
- **TLEN** = 203 (template/insert length in bp)

3. Optional tags (after QUAL, written TAG:TYPE:VALUE, type Z = text, i = integer)

Tag	Meaning	Our read
MC:Z	CIGAR of the mate	100M
MD:Z	Mismatch string: number of matching bases, reference base at a mismatch	100 (all 100 bases match)
PG:Z	Program that last wrote the read (ID of an @PG line)	MarkDuplicates
RG:Z	Read group (ID of the @RG line)	HWI-ST466.C1TD1ACXX.normal
NM:i	Edit distance to the reference	0
MQ:i	Mapping quality of the mate	60
AS:i	Alignment score (bwa)	100
XS:i	Score of the best other hit (bwa)	0

Reading this read

100M, MD:Z:100, NM:i:0 and AS:i:100 all say the same: the 100 bases match the reference. MAPQ 60 and XS:i:0 say the placement is unique.

Resource

Cancer Variant Analysis - SIB

Learn more at the SIB course:

- NGS - Quality Control, Alignment, Visualization
- https://broadinstitute.github.io/picard/explain-flags.html

Marking PCR Duplicates

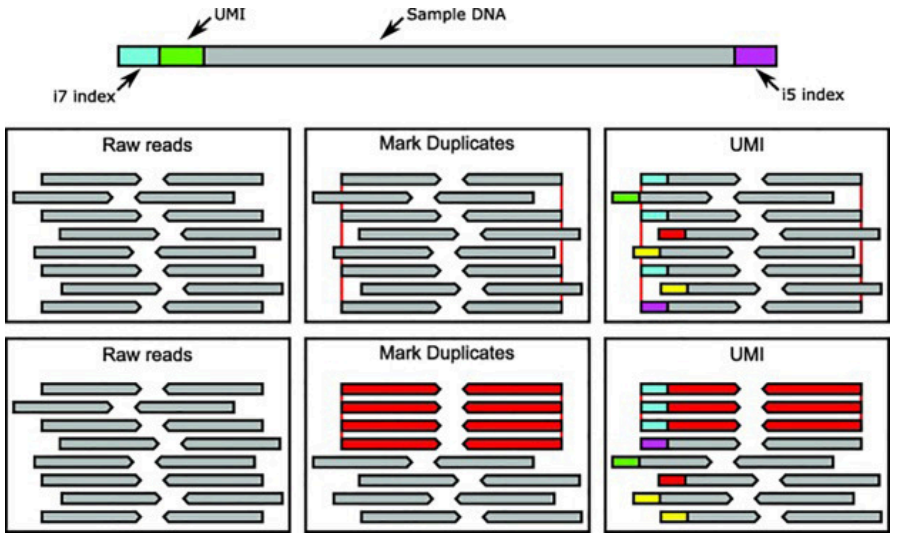
Why it matters:

- Variant callers assume each read is an **independent observation**
- PCR/optical duplicates violate this assumption
- Can lead to **false positive** variant calls

Solution:

- Mark duplicates based on alignment coordinates
- Use **Unique Molecular Identifiers (UMIs)** for accurate deduplication - especially important for low-input and ctDNA samples

Tool: `gatk MarkDuplicates`



Mark duplicates, Source

Read Groups: Organizing Your Data

Purpose: Track metadata for groups of reads within BAM files

Key Read Group Tags

Tag	Meaning	Good to know
`ID`	Unique read group identifier	
`SM`	**Sample name** (patient/specimen)	Critical for multi-sample calling. Tumor vs. Normal distinction
`LB`	Library prep identifier	Used for duplicate marking
`PL`	Sequencing platform	
`PU`	Platform unit (flowcell.lane)	For tracking batch effects

Why read groups matter

Step	What the read group does
BQSR	Models built per-read-group
Merging	Combine multiple lanes/runs per sample
Duplicate marking	Per-library detection
Somatic calling	Required to distinguish tumor/normal
QC & troubleshooting	Track technical artifacts

Alignment Record with Read Group:

Tag	Value	Meaning
RG:Z:	`tumor.lane1`	Links read to `@RG ID:tumor.lane1`

Header Example (Tumor-Normal Pair)

Normal:

```
@RG  ID:normal.lane1  SM:patient001_normal  LB:lib_n  PL:ILLUMINA  PU:HWI.lane1
@RG  ID:normal.lane2  SM:patient001_normal  LB:lib_n  PL:ILLUMINA  PU:HWI.lane2
```

Tumor:

```
@RG  ID:tumor.lane1   SM:patient001_tumor   LB:lib_t  PL:ILLUMINA  PU:HWI.lane1
@RG  ID:tumor.lane2   SM:patient001_tumor   LB:lib_t  PL:ILLUMINA  PU:HWI.lane2
```

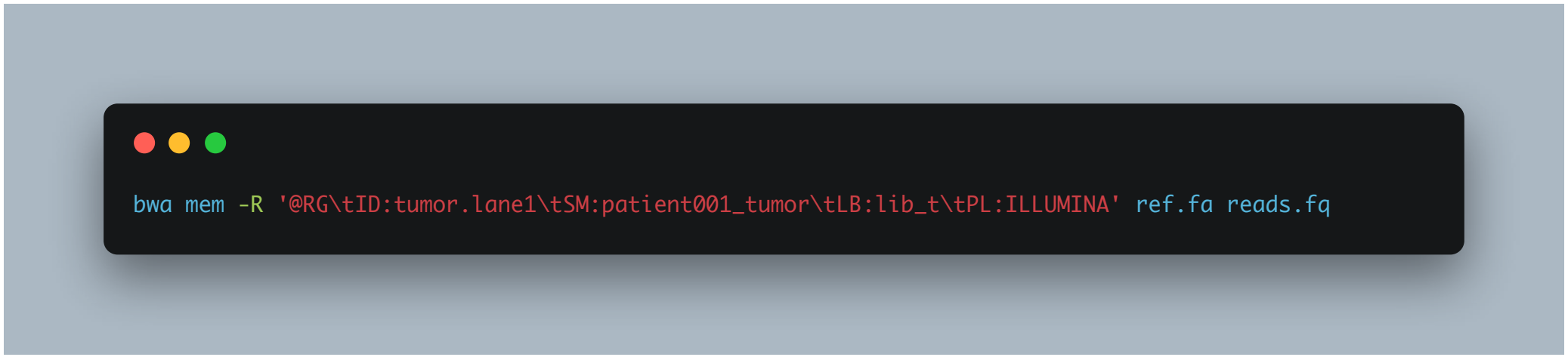
```
read1  99  chr20  1000  60  76M  =  1150  226  ATGC...  ####  RG:Z:tumor.lane1
```

Somatic Variant Calling Context

Mutect2 uses the `SM:` field to identify tumor and normal samples. Without proper read groups, somatic callers cannot distinguish between samples!

Best Practice

Add read groups **during alignment** for efficiency:



Avoid post-hoc addition (e.g., GATK `AddOrReplaceReadGroups`) when possible.

Somatic Variant Calling Challenges

Germline calling assumes:

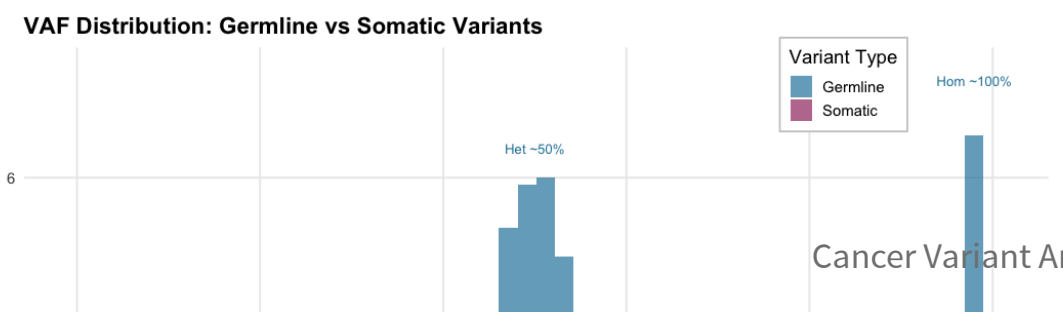
- Heterozygous: ~50% VAF (typically 30-70% due to technical variation)
- Homozygous: ~100% VAF

These assumptions fail in tumors:

- Variable tumor purity (30-80%, which is effectively contamination by normal cells)
- Clonal heterogeneity (subclones)
- Copy number alterations
- **VAF can be anywhere from <1% to 100%**

Additional challenges:

- Sequencing errors, alignment artifacts
- FFPE artifacts
  - C>T/G>A deamination
  - 8-oxoG which causes the G>T transversions
- Sample contamination



Let's make an example (assuming diploid genome and CN=2)

1. The Ideal Case
Sample is 100% Tumor

If a mutation is heterozygous (1 of 2 alleles):

$$\text{VAF} = \frac{1}{2} = \mathbf{50\%}$$

To a caller, this looks like a clear, standard germline variant.

2. Closer to reality
Sample is 40% Tumor (60% Normal)

The normal cells (wild type) dilute the signal.

$$\text{VAF} = \frac{\text{Purity}}{2}$$

$$\text{VAF} = \frac{0.40}{2} = \mathbf{20\%}$$

⚠ **Real-world complications**

Aneuploidy complicates this further: In real cases, copy number alterations can dramatically change expected VAF. For example, if the mutation is on a region with CN=4, the math becomes more complex.

Tumor-infiltrated controls: Matched control samples can also be tumor-infiltrated, making it sometimes impossible to distinguish true somatic variants from germline or contamination.

Tumor Purity, Ploidy, and Clonality

Key concepts:

Tumor purity: Fraction of tumor cells in sample

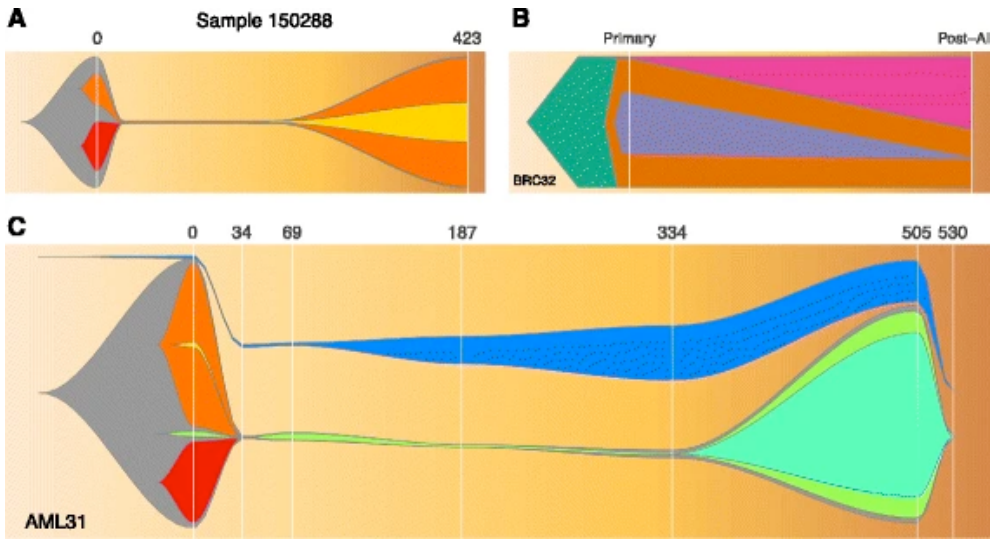
Ploidy: Average copy number across genome (often >2 in cancer)

Clonal variants: Present in all tumor cells

Subclonal variants: Present in a subset of tumor cells

VAF interpretation examples:

- Pure tumor (100%), clonal het variant → VAF ≈ 50%
- 50% purity, clonal het variant → VAF ≈ 25%
- 50% purity, subclonal variant (20% of cells) → VAF ≈ 5%



Fish plot of clonal evolution: Panel a A case of primary and relapsed AML. Panel b A breast cancer before and after neoadjuvant aromatase inhibitor therapy. Panel c An AML with complex clonal structure and 7 timepoints, Source

Clinical Relevance

Subclonal variants can become dominant after treatment selection pressure, leading to resistance.

Variant Filtering Strategies

Three key considerations:

1. Sequencing Error

- Base quality scores (Phred: Q30 = 1/1000 error)
- Variant allele frequency
- Strand bias
- Mapping quality (ie MAPQ ≥ 20)

2. Technical Artifacts

- **Panel of Normals (PoN)**: Database of artifacts seen in normal samples
- Systematic errors from library prep or sequencing
- FFPE artifacts

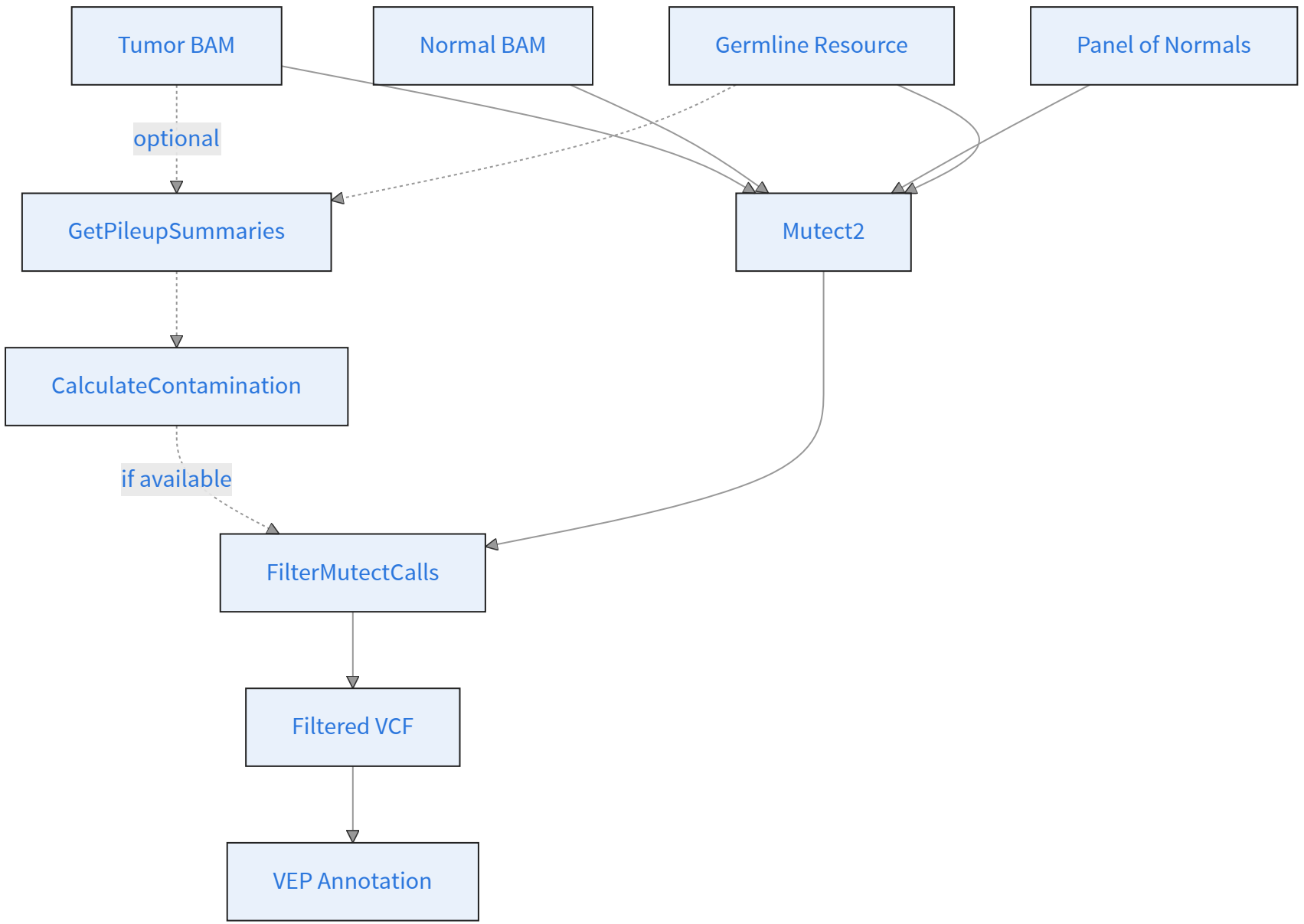
3. Germline Filtering

- Compare with matched normal
- Filter using population databases:
  - **gnomAD v4.0** (>800K individuals)
  - 1000 Genomes Phase 3 (~2,500 individuals)
- Common variants (AF > 0.1%) are typically germline

gnomAD

The Genome Aggregation Database contains variant frequencies essential for filtering common germline variants. Most callers (Mutect2, Strelka2) apply these filters automatically, but understanding the logic helps with troubleshooting false positives.

GATK Mutect2 Workflow



Key features:

- Haplotype-aware variant calling (local assembly)
- Joint analysis of tumor-normal pairs
- Integrated contamination estimation
- F1R2 (Forward 1st, Reverse 2nd) artifact filtering

> **PoN vs gnomAD**
>
> - **Germline Resource**: Filters biological germline variants (universal)
> - **PoN**: Filters technical artifacts (platform-specific)
>
> Public PoNs exist but work best when your sequencing protocol matches theirs. When in doubt, build your own!

You can find every step's relative script here

The VCF File Format

Variant Call Format

Standard for storing variant data (current: v4.3)

```
##fileformat=VCFv4.3
##INFO=<ID=DP,Number=1,Type=Integer,Description="Total Depth">
##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency">
##FILTER=<ID=weak_evidence,Description="Insufficient support">
##FILTER=<ID=germline,Description="Likely germline variant">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=AD,Number=R,Type=Integer,Description="Allelic depths (REF,ALT)">
##FORMAT=<ID=AF,Number=A,Type=Float,Description="Allele Frequency">
#CHROM  POS     ID        REF  ALT  QUAL  FILTER  INFO        FORMAT      TUMOR         NORMAL
chr17   7577538 rs123     G    A    .     PASS    DP=100      GT:AD:AF    0/1:70,30:0.30  0/0:50,0:0.0
chr17   7578406 .         C    T    .     germline DP=85     GT:AD:AF    0/1:40,45:0.53  0/1:30,25:0.45
```

Key columns explained

Column	Meaning
`CHROM` / `POS`	Genomic location
`REF` / `ALT`	Reference and alternate alleles
`FILTER`	Quality filters applied (`PASS` = passed all)
`FORMAT`	Defines sample-specific fields
`GT`	Genotype (`0/0` = hom ref, `0/1` = het, `1/1` = hom alt)
`AD`	Allele depths (REF count, ALT count)
`AF`	Variant Allele Frequency = ALT/(REF+ALT)

Example interpretation

Row	Tumor	Normal	Verdict
1	`0/1:70,30:0.30`	`0/0:50,0:0.0`	Somatic mutation (30% VAF in tumor, 0% in normal) ✓
2	`0/1:40,45:0.53`	`0/1:30,25:0.45`	Germline variant (present in both tumor and normal) ✗

FILTER Field

"PASS" means the variant passed all filters, **not** that it's biologically validated. Clinical decisions require orthogonal validation (Sanger, ddPCR, amplicon-seq).

Variant Annotation: The Critical Step

After calling, variants need biological context:

Functional Annotation

- **VEP** (Ensembl) or **SnpEff**
- Effect prediction: missense, nonsense, splice site
- Impact assessment: SIFT, PolyPhen, CADD scores

Population Frequencies

- gnomAD, dbSNP
- Filter common germline variants

Clinical Databases

- **ClinVar**: Clinical significance
- **COSMIC**: Cancer mutation database
- **OncoKB**, **CIViC**: Actionability

> **Tools**
>
> VEP offers extensive plugin ecosystem.

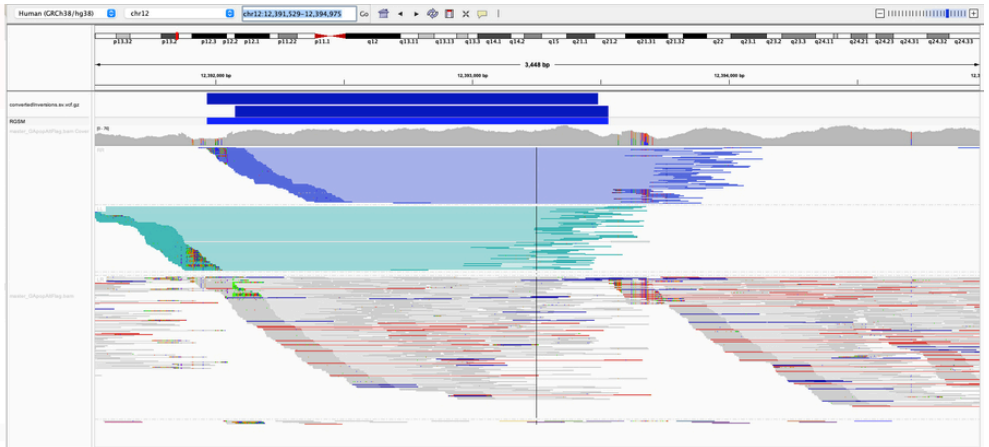
Structural Variation Detection

Types:

- Large insertions/deletions
- Translocations
- Inversions
- Complex rearrangements

Detection methods:

- Discordant read pairs
- Split reads
- Read depth changes
- Assembly-based approaches



SV: inversion in IGV, Source

Tools: Manta, Tiddit, GRIDSS2, DELLY, Sniffles2 (long reads)

> **Long Reads Advantage**
>
> Long-read sequencing (PacBio HiFi, ONT) dramatically improves structural variant detection, especially for complex rearrangements and insertions.

Copy Number Variation (CNV)

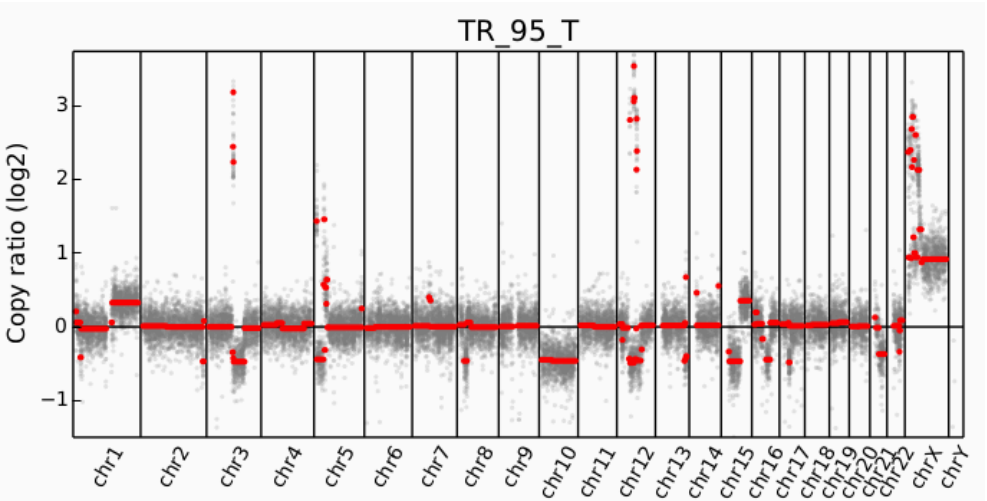
Characteristics:

- Gains or losses of genomic segments
- Full chromosome or arm-level events common
- Can cause Loss of Heterozygosity (LOH)

Detection approach:

1. Calculate coverage in bins
2. Normalize for GC content and mappability
3. Compare tumor vs normal ratio
4. Segment and call CNV regions

Tools: CNVkit, ASCAT, Control-FREEC, PURPLE



CNV scatter pot, Source

Clinical Example

ERBB2 (HER2) amplification in breast cancer determines eligibility for trastuzumab. MYC amplification is prognostic in many cancers.

Gene Fusions in Cancer

Mechanism:

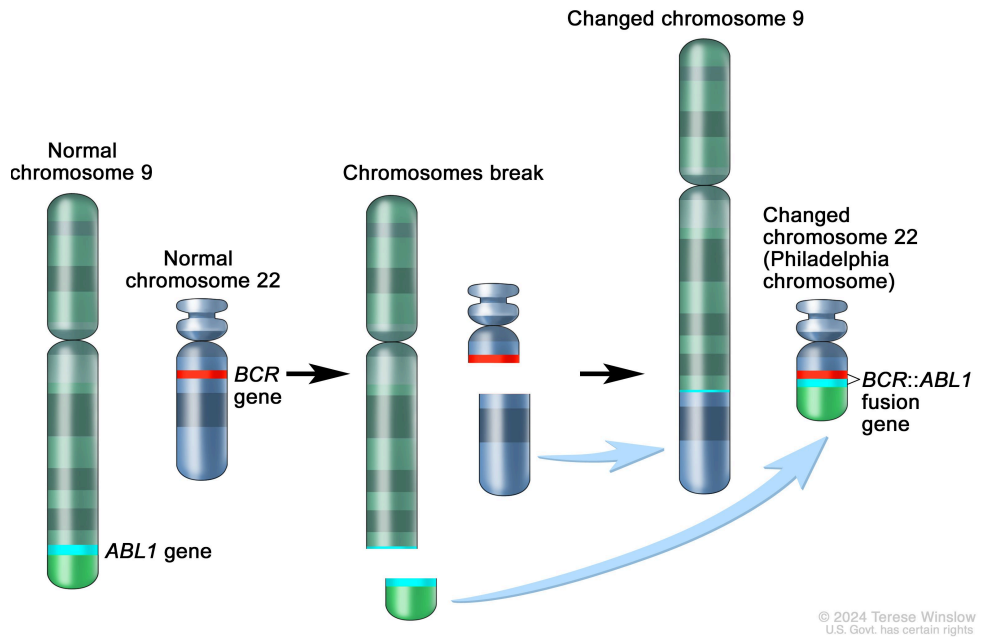
- Chromosomal translocation
- Fusion of gene elements
- Creates chimeric transcripts

Detection data types:

- WGS (genomic breakpoints)
- WES (if breakpoints in exons)
- RNA-seq (fusion transcripts)

Detection method: Discordant alignments where paired reads map to different genes/chromosomes

Tools: Manta (DNA), STAR-Fusion, Arriba (RNA-seq)



BCR::ABL1 fusion, Source

Famous Example

BCR-ABL fusion in CML(Chronic Myeloid Leukemia)

- discovered 1960
- translocation identified 1973
- imatinib approved 2001.

From observation to targeted therapy took 41 years; today we can do this computationally.

Quality Control Metrics

Essential QC checks at each stage:

Metric	Expected Value	Interpretation
Mapping rate	>95%	Low = contamination or poor quality
Duplicate rate	<30% (WGS)	High = low library complexity
Mean coverage	As specified	Low = insufficient data
Coverage uniformity	CV <0.2	High variability = capture issues
Contamination	<1-2%	High = sample swap or cross-contamination
Ti/Tv ratio	~2.0-2.1 (WGS), ~3.0-3.3 (exome)	Low = sequencing errors enriched
Insert size	200-400bp (Illumina)	Bimodal = library issues
GC bias	Flat across 30-70% GC	Strong bias = PCR or coverage issues

Tools: `fastQC`, `MultiQC`, `mosdepth`, `GATK CollectHsMetrics`

Summary and Key Takeaways

Cancer Genomics Fundamentals:

- Cancer is driven by genomic alterations
- Both small variants and structural changes
- Somatic vs germline distinction critical
- ~5-10% hereditary predisposition

Technical Considerations:

- Tumor-normal paired design
- Appropriate sequencing strategy
- Quality preprocessing essential
- Purity >30% recommended

Variant Calling:

- Haplotype-aware methods (Mutect2)
- Multiple filtering strategies
- Standard file formats (BAM, VCF)
- **Annotation is essential!**

Clinical Applications:

- Mutational signatures reveal etiology
- TMB predicts immunotherapy response
- Fusions guide targeted therapy
- CNV determines treatment options

References and Resources

Key Papers:

- Hanahan & Weinberg (2011) Cell - Hallmarks of Cancer
- Vogelstein et al. (2013) Science - Cancer Genome Landscapes
- Alexandrov et al. (2020) Nature - Mutational Signatures
- Cibulskis et al. (2013) Nature Biotech - MuTect
- Karczewski et al. (2020) Nature - gnomAD

Databases:

- COSMIC (cancer.sanger.ac.uk) - Somatic mutations
- gnomAD (gnomad.broadinstitute.org) - Population frequencies
- ClinVar - Clinical interpretations
- OncoKB - Precision oncology knowledge

Tools & Pipelines:

- GATK (gatk.broadinstitute.org)
- nf-core/sarek - Production pipeline
- IGV (igv.org) - Visualization

Exercises



Giphy